LAB OBSERVATION

EXPERIMENT-15

15) Develop a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.

Answer)

Aim:

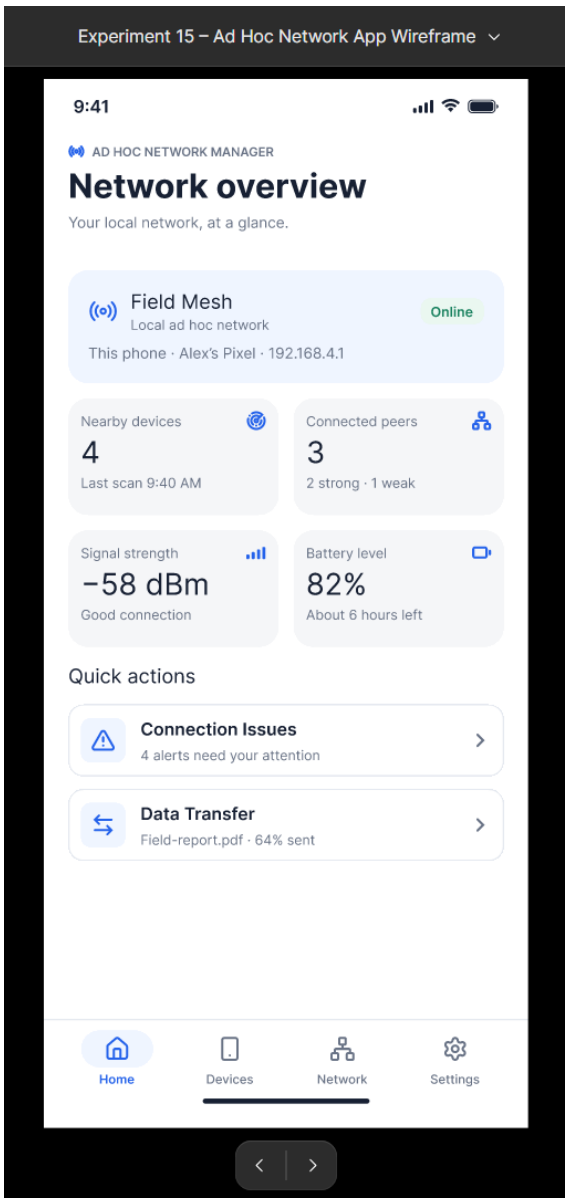
To develop a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.

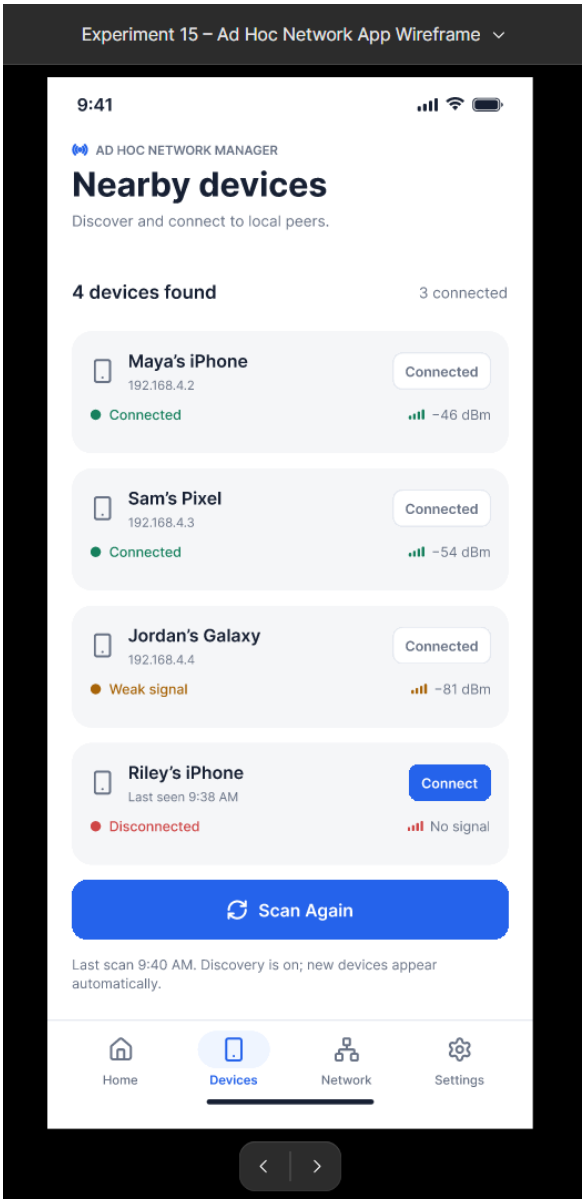
Procedure:

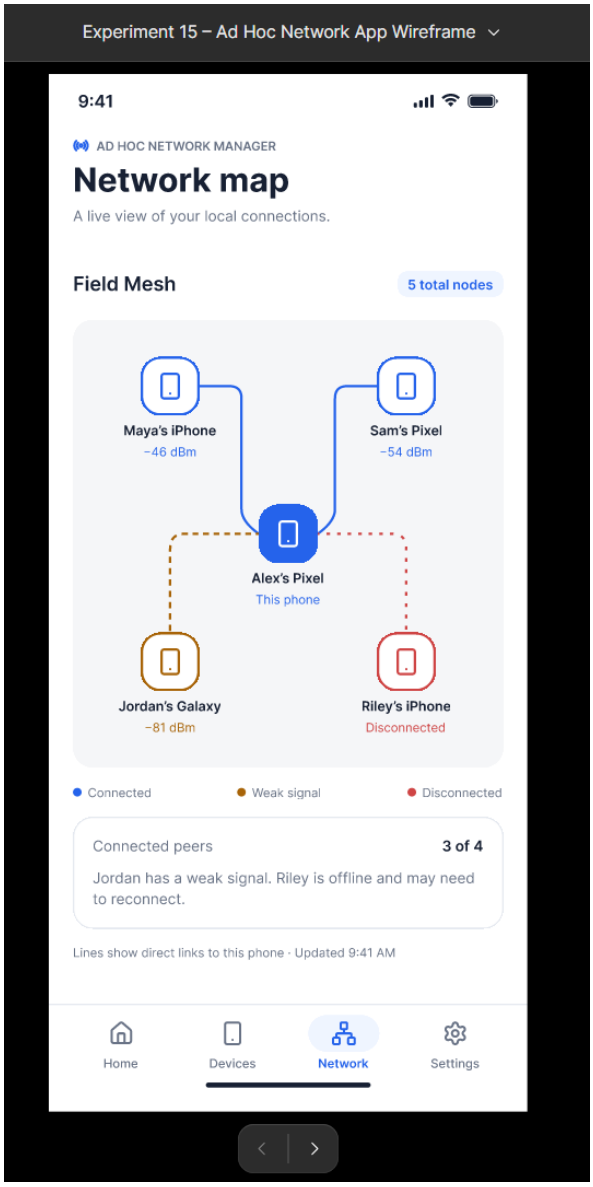
1. Open Figma.
2. Create a new design file.
3. Select a suitable mobile frame of size 390 × 844 pixels.
4. Design the home dashboard displaying network status, nearby nodes, signal strength, and battery level.
5. Create a screen to display nearby devices and connection options.
6. Design a network map showing connected devices and their communication links.
7. Add a connection issues screen with alerts for weak signals, disconnected nodes, and network congestion.
8. Design a data transfer screen and settings screen with suitable options.
9. Add icons, colors, labels, and navigation buttons to improve usability.
10. Connect the screens using prototype interactions, review the design, and save the Figma file.

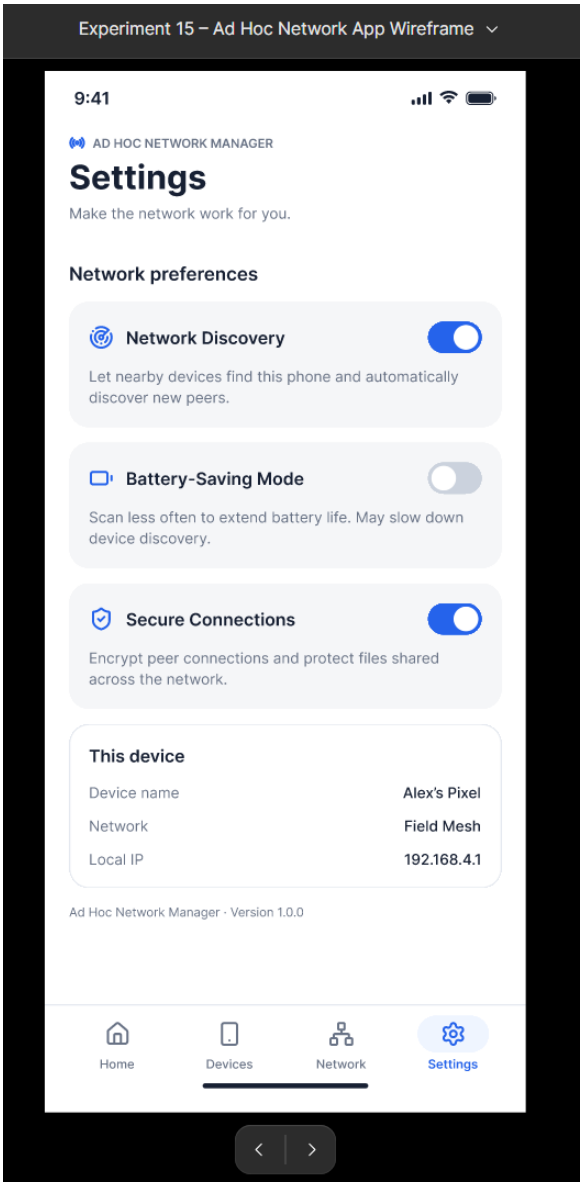
Output:

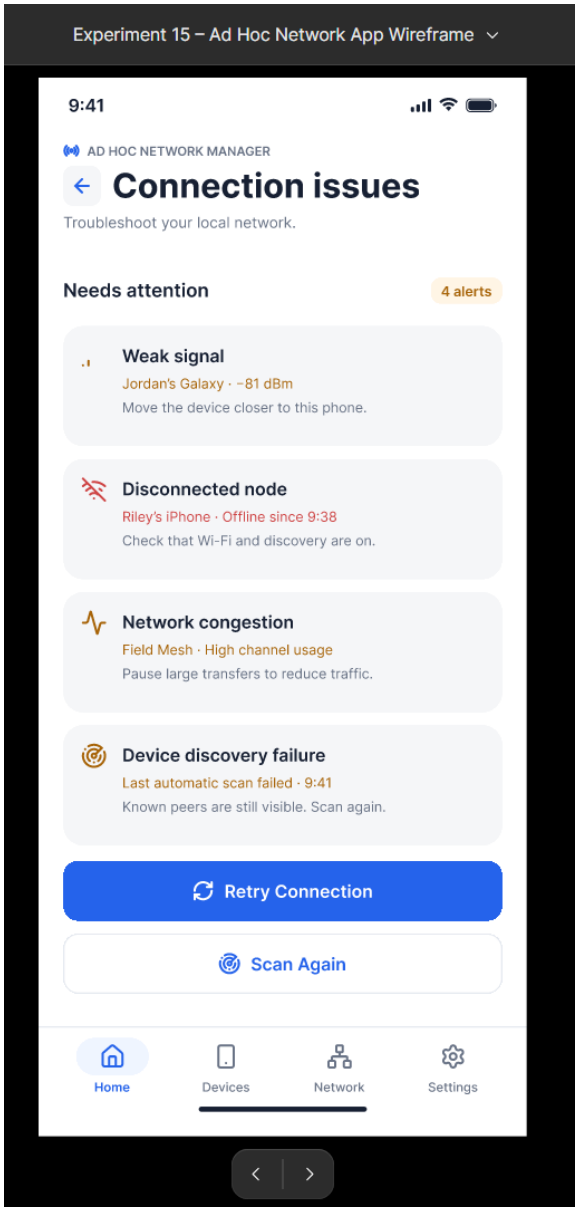
A mobile application wireframe displaying network status, nearby devices, network connections, connection issue alerts, data transfer, and settings for an ad hoc wireless network.

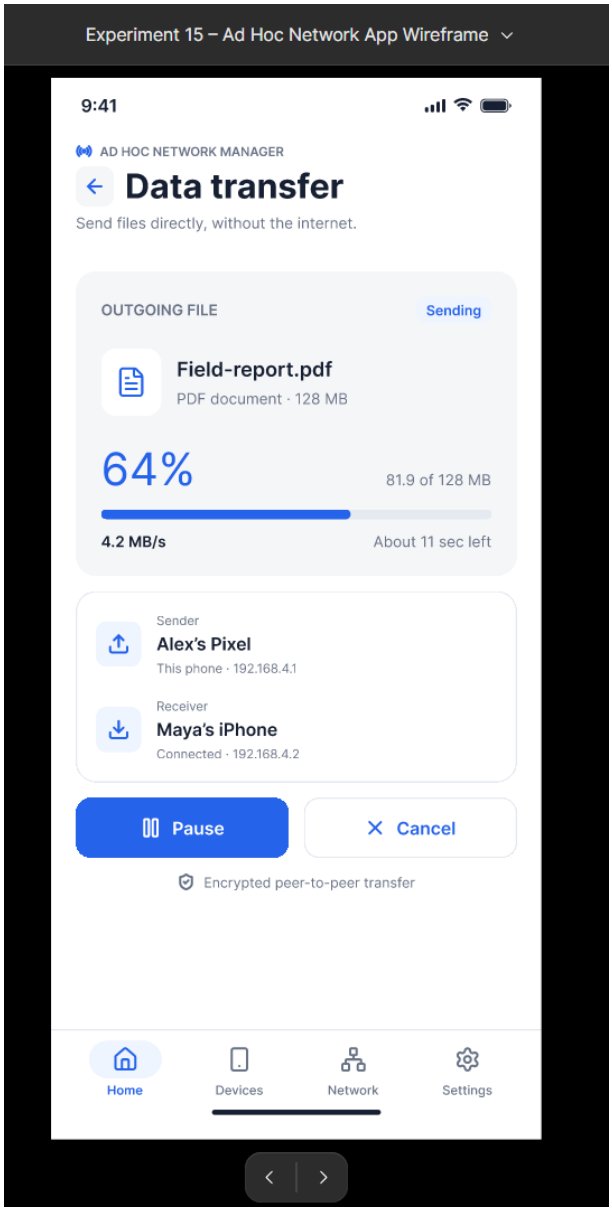

Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Network overview
Your local network, at a glance.
Field Mesh
Local ad hoc network
Online
This phone · Alex's Pixel · 192.168.4.1
Nearby devices
4
Last scan 9:40 AM
Connected peers
3
2 strong · 1 weak
Signal strength
−58 dBm
Good connection
Battery level
82%
About 6 hours left
Quick actions
Connection Issues
4 alerts need your attention
Data Transfer
Field-report.pdf · 64% sent
Home
Devices
Network
Settings


Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Nearby devices
Discover and connect to local peers.
4 devices found
3 connected
Maya's iPhone
192.168.4.2
Connected
Connected
−46 dBm
Sam's Pixel
192.168.4.3
Connected
Connected
−54 dBm
Jordan's Galaxy
192.168.4.4
Connected
Weak signal
−81 dBm
Riley's iPhone
Last seen 9:38 AM
Connect
Disconnected
No signal
Scan Again
Last scan 9:40 AM. Discovery is on; new devices appear automatically.
Home
Devices
Network
Settings

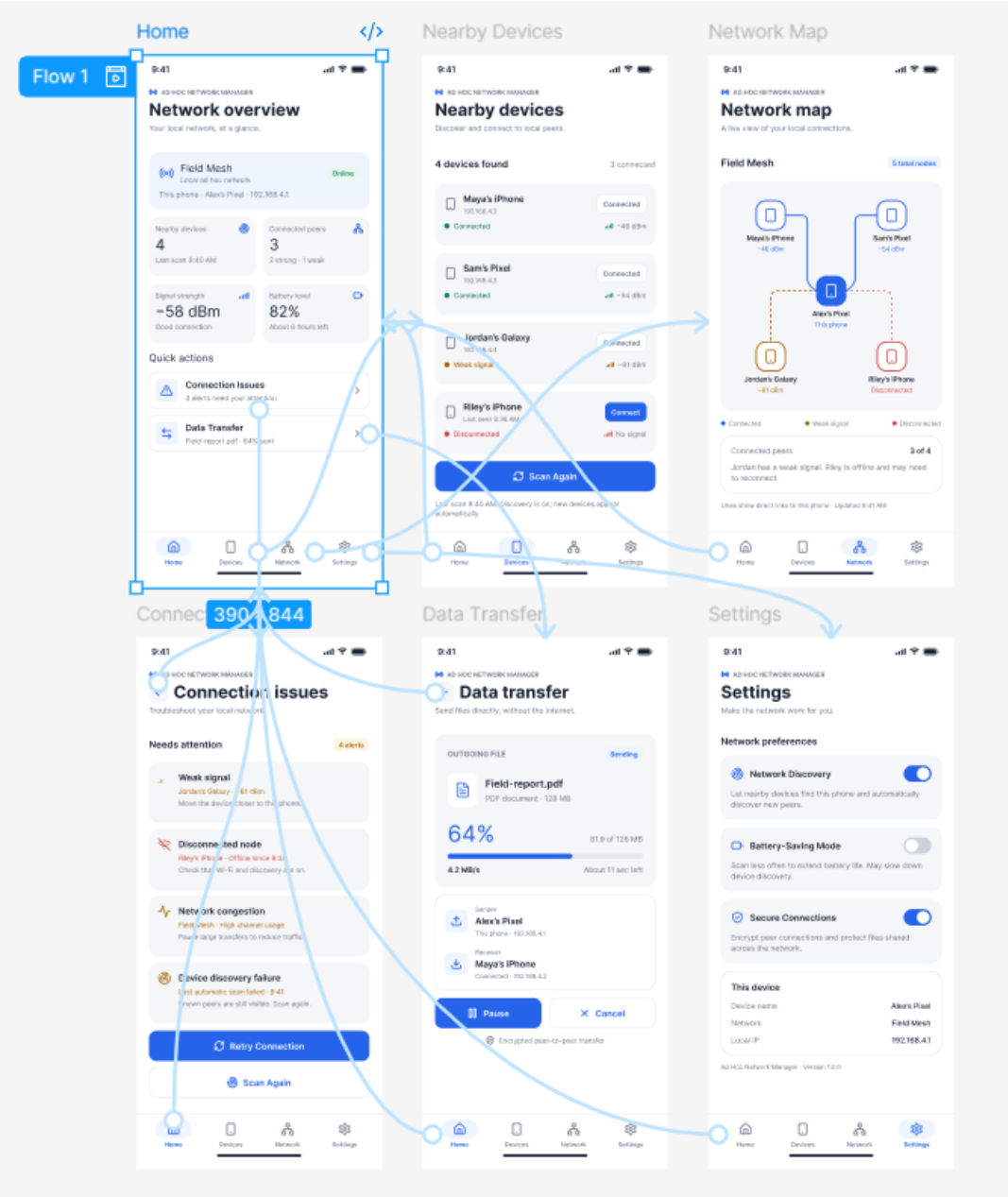

Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Network map
A live view of your local connections.
Field Mesh
5 total nodes
Maya's iPhone
−46 dBm
Sam's Pixel
−54 dBm
Alex's Pixel
This phone
Jordan's Galaxy
−81 dBm
Riley's iPhone
Disconnected
Connected
Weak signal
Disconnected
Connected peers
3 of 4
Jordan has a weak signal. Riley is offline and may need to reconnect.
Lines show direct links to this phone · Updated 9:41 AM
Home
Devices
Network
Settings


Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Settings
Make the network work for you.
Network preferences
Network Discovery
Let nearby devices find this phone and automatically discover new peers.
Battery-Saving Mode
Scan less often to extend battery life. May slow down device discovery.
Secure Connections
Encrypt peer connections and protect files shared across the network.
This device
Device name
Alex's Pixel
Network
Field Mesh
Local IP
192.168.4.1
Ad Hoc Network Manager · Version 1.0.0
Home
Devices
Network
Settings


Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Connection issues
Troubleshoot your local network.
Needs attention
4 alerts
Weak signal
Jordan's Galaxy · −81 dBm
Move the device closer to this phone.
Disconnected node
Riley's iPhone · Offline since 9:38
Check that Wi-Fi and discovery are on.
Network congestion
Field Mesh · High channel usage
Pause large transfers to reduce traffic.
Device discovery failure
Last automatic scan failed · 9:41
Known peers are still visible. Scan again.
Retry Connection
Scan Again
Home
Devices
Network
Settings


Experiment 15 – Ad Hoc Network App Wireframe
9:41
AD HOC NETWORK MANAGER
Data transfer
Send files directly, without the internet.
OUTGOING FILE
Sending
Field-report.pdf
PDF document · 128 MB
64%
81.9 of 128 MB
4.2 MB/s
About 11 sec left
Sender
Alex's Pixel
This phone · 192.168.4.1
Receiver
Maya's iPhone
Connected · 192.168.4.2
Pause
Cancel
Encrypted peer-to-peer transfer
Home
Devices
Network
Settings

Implementation:



Result:

The wireframe for a mobile application addressing challenges related to ad hoc wireless networks was successfully designed in Figma.